Supplement to "Multi-armed bandits with arriving arms: sequential screening and reliable set maintenance"

October 15, 2026

Notation and scope

The observation model has independent arm streams, each consisting of independent, identically distributed rewards in $[0,1]$, deterministic permanent arrivals $\mathcal{A}_t$, and one observation per physical time step. Let $r^*(t) = \max_{i\in\mathcal{A}_t} r_i$, $\Delta_i(t) = r^*(t) - r_i$, and $\mathcal{G}_t^\varepsilon = \{i : \Delta_i(t) \le \varepsilon\}$. Reports are issued after the current observation and partition $\mathcal{A}_t$ as $(\mathcal{P}_t, \mathcal{N}_t, \mathcal{V}_t)$. Write

$$E_{\text{lab}} = \{\forall t : \mathcal{P}_t \subseteq \mathcal{G}_t^\varepsilon \subseteq \mathcal{A}_t \setminus \mathcal{N}_t\}, \qquad \mathcal{U}_T(\eta) = \sum_{t\le T}\sum_{i\in\mathcal{A}_t} \mathbf{1}\{i \in \mathcal{V}_t, |\Delta_i(t) - \varepsilon| > \eta\},$$

and $R_T = \sum_{t\le T} \Delta_{I_t}(t)$. Reliability means $P(E_{\text{lab}}) \ge 1-\delta$ for every allowed environment. $\mathfrak{R}_\Theta(a)$ minimizes worst-case ER_T over full-model reliable policies with $P_\theta(E_{\text{lab}}, \mathcal{U}_T \le a) \ge 1-\delta$ for all $\theta \in \Theta$. A bar denotes averaging under the independent uniform prior on all candidate signs. These priors are proof devices and impose no Bayesian requirement on the policy.

All probability and sample-budget interfaces are proved below. Classical change of measure and multiple-recovery scales are credited to the existing literature (Kaufmann et al., 2016; Malloy and Nowak, 2014; Jamieson and Jain, 2018; Mason et al., 2020).

S1 A fixed-budget analysis of the screening-based UCB-AA procedure

S1.1 Exact scope and executable conventions

Fix a deterministic physical horizon $T \ge 1$ and $0 < \delta < 1$. There are finitely many available arms by every finite time, at least one arm is available at time one, and arrivals are deterministic and permanent. The arm streams are independent, and each consists of iid $[0,1]$-valued rewards with mean r_i. The proof also applies conditionally on an independent exogenous arrival schedule. No gap-stability or cumulative-jump assumption is imposed. The learner observes the arrivals before selecting the arm at each physical time. Newly arriving arms are buffered until the next algorithmic round. All observations are retained, and an eliminated arm is not sampled again.

Set

$$M_T = \left\lceil \frac{1}{2}\log_2(T+1) \right\rceil, \qquad N = \frac{8T4^{M_T}}{\delta}, \qquad \widetilde{\Delta}_p = 2^{-p}, \tag{1}$$

and, for $0 \le p \le M_T$, retain the original target and radius formulas

$$n_p = \left\lceil \frac{2\log(N\widetilde{\Delta}_p^2)}{\widetilde{\Delta}_p^2} \right\rceil, \qquad b_p = \sqrt{\frac{\log(N\widetilde{\Delta}_p^2)}{2n_p}}. \tag{2}$$

The calibration in (1) is a sufficient choice for this theorem; the smaller empirical calibration in the submitted version is not covered by this probability guarantee.

Write s_m for the physical starting time of algorithmic round m, with $s_0 = 1$. At s_m admit every arm in the buffer, including arrivals at s_m. The admitted batch is A_m, its cumulative union is $A(m)$, the incumbent survivors are C_{m-1}, and the initial active set is $B_m = C_{m-1} \cup A_m$. Put $C_{-1} = \varnothing$. These are the round-based sets of the original UCB-AA procedure. Its notation $K(m)$, when used, denotes a prescribed nondecreasing upper bound on $|A(m)|$; it is different from the actual physical-time count $K_t = |\mathcal{A}_t|$. This proof does not require $K(m)$ as an additional input because the calibration controls at most T observed streams.

The following conventions make the original two-stage architecture precise. They are part of the analyzed procedure, not conclusions about unspecified implementations.

1. An elimination call at phase p first samples every arm in its candidate set S until its lifetime count reaches n_p. An arm already at or above that target receives no additional sample. After the entire call's sampling is complete, form

   $$\mathrm{LCB}_i(p) = \hat{r}_i(n_p) - b_p, \qquad \mathrm{UCB}_i(p) = \hat{r}_i(n_p) + b_p$$

   from the actual first n_p observations of each reference arm. Remove simultaneously from S exactly those arms for which

   $$\mathrm{UCB}_i(p) < \max_{j \in R} \mathrm{LCB}_j(p).$$

   In particular, the inequality is strict. Prefix statistics for incumbents are retained; an unobserved prefix is never substituted into a comparison.

2. For $m \ge 2$, run preliminary phases $p = 0, \ldots, m-2$ on the surviving part of A_m, with reference set equal to that part together with C_{m-1}. Only new arms are removed in these calls. Then run formal phases $p = m-1, m$ on the merged survivors. For $m = 1$, run only formal phases $0, 1$; for $m = 0$, run formal phase 0.

3. If a new batch is empty, all its preliminary phases are no-ops: the set entering formal phase $m-1$ is C_{m-1}. An empty surviving new batch is handled the same way. This specifies the missing intermediate set in the original pseudocode when preliminary screening is skipped.

4. Sampling orders and ties are fixed by observed history and independent policy randomization. Every physical pull counts toward T. Stop immediately after pull T, even in the middle of

an arm's target, an elimination call, or a round. No terminal round is appended to remove buffered arrivals from the regret ledger.

Thus the alteration preserves buffering, screening against incumbent evidence, and formal elimination. It repairs the statistical bookkeeping and stopping convention; it does not replace UCB-AA by a reward-service mixture or by CP-UCB-AA.

S1.2 Three distinct contributions to regret

At a physical time $t \le T$, let $m(t)$ be its current algorithmic round. Define the best initially active mean and the best admitted mean by

$$r_{*(m)} = \max_{i \in B_m} r_i, \qquad r_{\mathrm{adm}}(m) = \max_{i \in A(m)} r_i.$$

For an admitted arm in round m, distinguish the original roundwise gap $\Delta_i^{\mathrm{rd}}(m) = r_{*(m)} - r_i$ from the physical-time gap $\Delta_i(t) = r^*(t) - r_i$, where $r^*(t) = \max_{i \in \mathcal{A}_t} r_i$. The superscript marks the algorithmic-round clock; it does not alter the physical-time gap definition. The following three truncated quantities have different meanings:

$$\begin{aligned} R_{\mathrm{round}}(T) &= \sum_{t=1}^{T} \{r_{*(m(t))} - r_{I_t}\}, \\ \mathcal{H}_T &= \sum_{t=1}^{T} \{r_{\mathrm{adm}}(m(t)) - r_{*(m(t))}\}, \\ D_T &= \sum_{t=1}^{T} \{r^*(t) - r_{\mathrm{adm}}(m(t))\}. \end{aligned} \tag{3}$$

Here $\mathcal{H}_T$ is the deficit caused by previously eliminated admitted arms, whereas D_T is arrival delay alone. On every sample path,

$$R_T(\pi) = R_{\mathrm{round}}(T) + \mathcal{H}_T + D_T. \tag{4}$$

All three terms are nonnegative, since the selected arm belongs to $B_{m(t)}$ and $B_{m(t)} \subseteq A(m(t)) \subseteq \mathcal{A}_t$. In particular, the residual after subtracting roundwise regret is not in general arrival delay alone.

For a prescribed complete run of $M+1$ algorithmic rounds, the original $T_0(\pi)$ denotes its actual, generally random number of pulls, and $R_{\mathrm{round}}(T_0(\pi)) = R_M(\pi)$. The identity (4) applies at that completed endpoint as a pathwise accounting statement. The fixed-budget theorem below does not replace a deterministic T by $\mathbb{E}T_0(\pi)$ or assert an unproved bound at an arbitrary stopping time.

Arrival delay has an exact exposure representation, including an unfinished last round. Let $\beta_u = r^*(u) - r^*(u-1) \ge 0$ for $u \ge 2$, and define

$$\alpha_T(u) = \min\big(\{s_m : u \le s_m \le T\} \cup \{T+1\}\big).$$

Because every available arm is admitted at a round boundary, $A(m) = \mathcal{A}_{s_m}$ and $r_{\mathrm{adm}}(m) = r^*(s_m)$.

Changing the order of two finite sums yields

$$D_T = \sum_{u=2}^{T} \beta_u \{\alpha_T(u) - u\}. \tag{5}$$

An improvement arriving exactly at a round boundary has zero delay. An improvement that is still buffered at T is charged through T.

Theorem S1.1 (UCB-AA at a fixed physical budget)**.** *Under the model and executable conventions above, let $K_T = |\mathcal{A}_T|$. With probability at least $1-\delta$, the best admitted mean is always represented in the active set, $\mathcal{H}_T = 0$, and*

$$R_T(\pi) \le \min\left\{T,\ 8\sqrt{(2\log N + 1)K_T T} + D_T\right\}. \tag{6}$$

Consequently

$$\mathbb{E}R_T(\pi) \le \min\left\{T,\ 8\sqrt{(2\log N + 1)K_T T} + \mathbb{E}D_T + \delta T\right\}. \tag{7}$$

The confidence event and both bounds include the last incomplete round and all samples in incomplete calls. The statement does not require a unique best arm and does not compress an elimination-time gap into a terminal gap.

Proof. First check the targets and the stopping convention. Since $\log(N4^{-M_T}) = \log(8T/\delta) > 1$, the function $x \mapsto 2x\log(N/x)$ is increasing on $[1, 4^{M_T}]$. More explicitly, its successive phase differences are

$$2\,4^{p-1}\{3\log(N4^{-p}) - \log 4\} > 1, \qquad p \ge 1.$$

Therefore n_p is strictly increasing. Also

$$b_p \le \widetilde{\Delta}_p/2, \qquad n_p \le (2\log N + 1)4^p, \qquad n_{M_T} > T. \tag{8}$$

The last fact follows from $4^{M_T} \ge T+1$ and $\log(N4^{-M_T}) > 1$. Formal elimination with a strict inequality never removes every arm, even off the confidence event: an arm with maximal lower confidence bound cannot satisfy the removal rule because its upper bound is at least its own lower bound. Initial and subsequent active sets are therefore nonempty. Completing round M_T would require a surviving arm to have $n_{M_T} > T$ observations, and hence cannot happen before the physical stopping time. The finite phase range suffices on every path.

For the global event, order arms by their first observed reward. At most T streams are exposed by time T. An arm's unobserved reward stream is independent of the history that first selects it, and the bounded iid Hoeffding inequality is uniform over its mean. For each exposed stream, the union of its potential-prefix failures over $p = 0, \dots, M_T$ has probability at most

$$\sum_{p=0}^{M_T} 2\exp(-2n_p b_p^2) = \frac{2}{N}\sum_{p=0}^{M_T} 4^p < \frac{8\,4^{M_T}}{3N}.$$

Here the potential prefix is defined even if the policy never observes that many rewards. Applying the same bound conditionally at the first exposure and summing over at most T exposed streams gives an event E with

$$\mathbb{P}(E^c) \le \frac{8T4^{M_T}}{3N} = \frac{\delta}{3} \le \delta, \qquad |\hat{r}_i(n_p) - r_i| \le b_p \tag{9}$$

for every observed arm and every phase needed in a comparison. One direct proof of the concentration inequality used here is as follows. For a random variable $Z \in [0,1]$, the second derivative of its centered log moment generating function is a tilted variance and is at most $1/4$. Twice integrating gives $\mathbb{E}e^{\lambda(Z-\mathbb{E}Z)} \le e^{\lambda^2/8}$. Multiplication over independent observations and exponential Markov inequality give the two-sided bound $2e^{-2nx^2}$. The exposure argument uses only independence of an unrevealed stream, not independence of its eventual sample count or admission round. More formally, let F_i be the union of potential-prefix failures for stream i, and let $q = (2/N)\sum_{p=0}^{M_T} 4^p$. Before revealing the reward at time t, condition on the observed history and the policy's current independent randomization. On the event that I_t has never been sampled, its stream is still unrevealed and the conditional probability of F_{I_t} is at most q. Summing over $t \le T$ bounds the probability that any first-exposed stream fails by Tq. This conditioning occurs before that stream's first reward is revealed.

Work on E. At every comparison, all reference means lie between their actual lower and upper confidence bounds. A maximal-mean arm in the reference set cannot satisfy the strict removal inequality. For formal elimination this preserves the maximum of the entire active set. In pre-elimination an incumbent maximum is not eligible for removal, and a new maximum cannot satisfy the removal inequality either. Induction over calls and rounds shows

$$\max_{i \in C_m} r_i = \max_{i \in A(m)} r_i$$

at every completed round, and the best admitted mean is preserved during every incomplete round as well. Thus $r_{*(m)} = r_{\mathrm{adm}}(m)$ and $\mathcal{H}_T = 0$.

There is no unavailable-prefix comparison in this induction. At the end of round $m-1$, every incumbent survivor has exactly n_{m-1} lifetime observations. In preliminary phase $p \le m-2$, this includes the stored first n_p observations. New reference arms have just been sampled to n_p before the simultaneous comparison. Formal calls likewise have the requisite prefixes after their sampling step.

Next establish the actual-count invariant. After any completed phase p in round m, every eligible arm that survives the comparison has

$$r_{*(m)} - r_i \le 4b_p. \tag{10}$$

Indeed a maximal-mean reference arm j is still present, so $\mathrm{LCB}_j(p) \ge r_{*(m)} - 2b_p$, whereas $\mathrm{UCB}_i(p) \le r_i + 2b_p$. Survival requires the latter quantity to be at least the former. This proves (10) without conditioning on a selected estimate. The claim concerns eligible new arms in a preliminary

call and all survivors in a formal call; no such bound is asserted for untested incumbents during preliminary screening.

Consider a physical pull that is the nth lifetime observation of its arm and occurs while filling target n_p in round m. If $p = 0$, its roundwise gap is at most one. If $p \geq 1$ and the arm is new in the current round, it survived phase $p-1$ of this same round before this pull, and its gap is at most $4b_{p-1}$. If it is an incumbent, no sample is needed in formal phase $m-1$, since it already has n_{m-1} observations. Its new samples occur only in phase m, after it survived the current round's formal phase $m-1$, giving the same bound. This includes $m = 1$; in round zero the only target is n_0. Consequently (8) gives, for every actual pull,

$$r_{*(m)} - r_i \leq \min\left\{1,\ 4\sqrt{\frac{2\log N + 1}{n}}\right\}. \tag{11}$$

For $p \geq 1$, use $4b_{p-1} \leq 4\widetilde{\Delta}_p$ and $n \leq n_p \leq (2\log N + 1)4^p$. For $p = 0$, use $n \leq n_0 \leq 2\log N + 1$. A partially sampled final phase requires only the preceding comparison, which has already finished, so the argument does not presume completion of the final target.

No arm's lifetime count is reset. Summing the distinct charges gives

$$\begin{aligned} R_{\text{round}}(T) &\leq 4\sqrt{2\log N + 1} \sum_{i:N_i(T)>0} \sum_{n=1}^{N_i(T)} n^{-1/2} \\ &\leq 8\sqrt{2\log N + 1} \sum_i \sqrt{N_i(T)} \leq 8\sqrt{(2\log N + 1)K_T T}. \end{aligned}$$

The second inequality follows by integrating $x^{-1/2}$ on each unit interval; the last is Cauchy–Schwarz and $\sum_i N_i(T) = T$. Combine this with (4), $\mathcal{H}_T = 0$, and the pathwise bound $R_T \leq T$ to obtain (6). Finally use $R_T \leq T$ on E^c, nonnegativity of D_T, and (9) to obtain (7). The expectation does not condition the law of selected rewards on coverage. □

For example, choosing $\delta = T^{-2}$ for $T \geq 2$ gives $\log N = O(\log(T+1))$. Thus the fixed-budget bound is sublinear whenever $K_T \log(T+1) = o(T)$ and $\mathbb{E}D_T = o(T)$. This is a sufficient arrival-delay condition stated on the physical timeline, not a guarantee of sublinear regret for arbitrary arrivals. The theorem is a corrected upper bound; no matching minimax claim follows merely from the square-root form of its statistical term.

S1.3 Static arms and fixed-gap arrivals

Corollary S1.2 (An initially available global optimum)**.** *Under Theorem S1.1, suppose an arm available at time one has mean $r^\star$, and no arm arriving by T has mean larger than $r^\star$. Put $\Delta_i = r^\star - r_i$ and $A = 2\log N + 1$. Then $D_T = 0$ on every sample path. With probability at least $1-\delta$,*

$$R_T(\pi) \le \min\left\{T,\ \sum_{\substack{i\in\mathcal{A}_T\\ \Delta_i>0}} \min\left\{\Delta_i(T-\tau_i+1),\ \frac{16A}{\Delta_i}\right\}\right\}. \tag{12}$$

The same deterministic sum bounds $\mathbb{E}R_T(\pi)$ after adding δT and taking the minimum with T.

Proof. The physical benchmark equals $r^\star$ throughout the horizon. Because an optimal arm is in the initial admitted batch, the best admitted mean also equals $r^\star$ on every path, even if that arm is later eliminated. Thus $D_T = 0$ pathwise. On the common coverage event, the active benchmark is $r^\star$ as well, so each pull's roundwise gap in (11) is its fixed physical gap Δ_i. If a suboptimal arm has $N_i(T) > 0$ observations, apply that inequality to its last observed prefix $n = N_i(T)$ to obtain

$$N_i(T) \le \frac{16A}{\Delta_i^2}.$$

This is an inequality for an integer count, so it requires no extra rounding term. It holds trivially at zero count. Independently, $N_i(T) \le T - \tau_i + 1$ by one pull per physical time and no prearrival sampling. The exact fixed-gap identity $R_T(\pi) = \sum_i \Delta_i N_i(T)$ gives (12); arms tied at the optimum contribute zero. On the exceptional event use $R_T \le T$. □

For $\delta = T^{-2}$ and $T \ge 2$, $A = O(\log(T+1))$. For a static finite arm set, this recovers the familiar upper order $O\{\sum_{i:\Delta_i>0} \log(T+1)/\Delta_i\}$. For arriving distractors satisfying $\Delta_i \ge g > 0$, it gives $O\{K_T \log(T+1)/g\}$, and hence $O(K_T \log(T+1))$ for fixed g. These are special-case upper bounds, not an assertion that preliminary screening is necessary for this order or a comparison proving linear regret for a procedure without preliminary screening.

The original fixed-gap illustration can be expressed on the exogenous physical clock: release an optimal arm of mean $3/4$ and an arm of mean $1/4$ at time one, and at deterministic physical times τ_i release arms of means $r_i = 1/2 - 2^{-i-2}$, $i \ge 1$. Their gaps are $1/4 + 2^{-i-2} \ge 1/4$, so the bound is $O(K_T \log(T+1))$. The release times here are external physical times, not algorithmic round indices chosen after seeing the procedure's random duration.

S1.4 A physical-time doubling wrapper

For completeness an unknown horizon can be handled by restarting at physical times $1, 2, 4, \ldots$. Epoch $\ell \ge 0$ has prescribed length $h_\ell = 2^\ell$ and local failure budget $\delta_\ell = \delta/(2h_\ell^2)$. At its start, initialize UCB-AA with all currently available arms as its initial batch, empty local counts, and the calibration (1) for horizon h_ℓ and failure budget δ_ℓ. Previously eliminated arms re-enter. Use only new rewards for this epoch's prefix statistics. Within the epoch, retain the two original stages and

buffer new arrivals until the next local round. Stop the last epoch at the actual physical endpoint T. This is a standard physical-budget restart, distinct from completing geometrically many algorithmic rounds regardless of their random duration.

Let D_T^{dbl} sum the arrival-delay terms of these epochs, with each epoch's unfinished local round truncated at its epoch endpoint or at T. Put

$$L_T = 2\log(128T^4/\delta) + 1.$$

Corollary S1.3 (A valid unknown-horizon extension)**.** *For every deterministic physical endpoint $T \geq 1$, the wrapper satisfies*

$$\mathbb{P}\left\{R_T(\pi) \leq 32\sqrt{L_T K_T T} + D_T^{\mathrm{dbl}}\right\} \geq 1 - \delta$$

and

$$\mathbb{E}R_T(\pi) \leq \min\left\{T,\ 32\sqrt{L_T K_T T} + \mathbb{E}D_T^{\mathrm{dbl}} + \delta\right\}.$$

The high-probability inequality can be made simultaneous over all physical endpoints on one event of probability at least $1 - \delta$.

Proof. Conditional on the history at an epoch start, future unrevealed rewards are independent arm streams, each consisting of iid rewards with its original mean. This follows by successively exposing the next unrevealed observation of a selected arm; selection and the epoch boundary depend only on already revealed entries. The fixed-budget confidence argument therefore applies conditionally, and its actual-prefix proof holds for every truncation of that epoch. The union of epoch failures has probability at most $\sum_{\ell \geq 0} \delta_\ell = 2\delta/3 < \delta$.

If $M_\ell = \lceil \frac{1}{2}\log_2(h_\ell + 1) \rceil$, then $4^{M_\ell} \leq 4(h_\ell + 1) \leq 8h_\ell$ and

$$N_\ell = \frac{8h_\ell 4^{M_\ell}}{\delta_\ell} \leq \frac{128h_\ell^4}{\delta}.$$

For every epoch that starts by T, its local $2\log N_\ell + 1$ is at most L_T. On the common good event, sum the fixed-budget bounds over complete epochs and the last partial epoch. Their available-arm counts are at most K_T. Since

$$\sum_{\ell=0}^{\lfloor \log_2 T \rfloor} \sqrt{2^\ell} \leq (2 + \sqrt{2})\sqrt{T} < 4\sqrt{T},$$

their statistical terms sum to at most $32\sqrt{L_T K_T T}$. This proves the simultaneous high-probability statement.

For the expectation apply the good-event bound separately within each epoch and charge at most h_ℓ regret for that epoch's failure. Earlier erroneous eliminations do not propagate through the

restart. The total exceptional-event contribution is at most

$$\sum_{\ell \geq 0} h_\ell \delta_\ell = \frac{\delta}{2} \sum_{\ell \geq 0} 2^{-\ell} = \delta.$$

Add the nonnegative expected delay terms and use $R_T \leq T$. □

Audit and boundaries of the repair. The proof uses only bounded-stream concentration, strict interval comparisons, the roundwise survivor-count invariant, a finite sum, and Cauchy–Schwarz; each dependency is proved above. The competing route of terminal-gap compression is unnecessary and is not reinstated. Ties, empty new batches, the first two rounds, singleton active sets, deterministic rewards, arbitrary within-round benchmark jumps, and an unfinished first or last target are all covered. The historical-deficit term vanishes only on the common confidence event. The delay term is retained explicitly, including at the endpoint. The unknown-horizon wrapper restarts the original screening algorithm and does not claim history retention across epochs. Neither result establishes that the original smaller empirical calibration has the same error probability, that the withdrawn gap-dependent corollary is true, or that the old round-based online theorem has been repaired without changing its clock.

S2 Proofs of the retained regret lower bounds

These arguments use the original regret criterion $R_T(\pi; \nu) = \sum_{t=1}^{T} (r^*(t) - r_{I_t})$ at a deterministic physical horizon. Reward laws are stationary and independent across arms and draws. The policies are nonanticipating, may use independent randomization, and use the same action kernels in every environment being compared. Each comparison keeps the deterministic arrival schedule fixed. The schedules and reward laws may depend on T where this is explicitly stated. All logarithms are natural.

S2.1 Information inequalities with adaptive sampling

For probability laws P, Q, write $D(P\|Q)$ for relative entropy and $\mathrm{TV}(P, Q) = \sup_A |P(A) - Q(A)|$. For Bernoulli laws the relative entropy is denoted by $\mathrm{kl}(p, q)$. The usual conventions apply at zero probabilities.

Lemma S2.1 (Information inequalities)**.** *For any event A,*

$$D(P\|Q) \geq \mathrm{kl}(P(A), Q(A)) \geq P(A) \log \frac{1}{Q(A)} - \log 2. \tag{13}$$

Also,

$$\mathrm{TV}(P, Q) \leq \sqrt{D(P\|Q)/2}, \qquad |\mathbb{E}_P Z - \mathbb{E}_Q Z| \leq \mathrm{TV}(P, Q) \quad (0 \leq Z \leq 1). \tag{14}$$

For $0 < q < 1$ and $0 \leq p \leq 1$,

$$\mathrm{kl}(p, q) \leq \frac{(p - q)^2}{q(1 - q)}. \tag{15}$$

Finally, suppose two bandit environments differ only in the reward law of arm i, from P_i to Q_i, with $D(P_i\|Q_i) < \infty$. If their history laws up to a deterministic time u are P^{H_u} and Q^{H_u}, then

$$D(P^{H_u}\|Q^{H_u}) = \mathbb{E}_P N_i(u) D(P_i\|Q_i), \qquad N_i(u) = \sum_{t=1}^{u} \mathbf{1}\{I_t = i\}. \tag{16}$$

Proof. Splitting the relative-entropy integral over A and its complement and applying Jensen's inequality on each part gives the first inequality in (13). The remaining binary entropy $p\log p + (1-p)\log(1-p)$ is bounded below by $-\log 2$, and $(1-p)\log(1/(1-q)) \geq 0$, giving the second inequality. For fixed $q \in (0,1)$, the second derivative of $p \mapsto \mathrm{kl}(p,q)$ is $1/(p(1-p)) \geq 4$; the function and its first derivative vanish at $p = q$. Thus $\mathrm{kl}(p,q) \geq 2(p-q)^2$, with endpoint cases by limits. Applying the first part of (13) to every event proves the total-variation bound in (14). For bounded Z,

$$\mathbb{E}_P Z - \mathbb{E}_Q Z = \int_0^1 \{P(Z > x) - Q(Z > x)\}\, dx,$$

so the same definition of total variation bounds the absolute difference. The inequality $\log x \leq x - 1$, applied to the two terms in Bernoulli relative entropy, gives (15).

To establish (16), include the observed rewards, actions, and available sets in the transcript. The available sets are deterministic and agree across the environments. Conditional on a past history h, the two joint laws of the next action and reward are $\pi_t(dj \mid h)P_j(dx)$ and $\pi_t(dj \mid h)Q_j(dx)$, respectively. The common action kernel cancels from the likelihood ratio. Their conditional relative entropy is therefore $\pi_t(\{i\} \mid h)D(P_i\|Q_i)$, since all other laws agree. The chain rule for relative entropy and iterated expectation give

$$D(P^{H_u}\|Q^{H_u}) = D(P_i\|Q_i) \sum_{t=1}^{u} \mathbb{E}_P \pi_t(\{i\} \mid H_{t-1}) = D(P_i\|Q_i)\mathbb{E}_P N_i(u).$$

This calculation also accounts for randomized policies. The finite horizon and finite armwise divergence ensure that its right side is finite. These are the standard adaptive change-of-measure identities used in bandit lower bounds (Kaufmann et al., 2016). □

S2.2 Inferior arrivals and the static specialization

For each T, let arm 0 be available from time 1 and uniquely optimal in a baseline environment $\nu^{(0)}$, with mean $r^\star$. All other means are strictly smaller. Let $\mathcal{I}_T$ be a nonempty finite set of arriving arms. For $i \in \mathcal{I}_T$, its law is P_i, its mean is $r^\star - \Delta_i$, and its arrival time is $\tau_i \leq T$. An alternative $\nu^{(i)}$ changes only this arm's law to Q_i, whose mean is $r^\star + \eta_i$. Put

$$H_i = T - \tau_i + 1, \qquad H_{\min} = \min_{i \in \mathcal{I}_T} H_i, \qquad d_i = D(P_i\|Q_i), \qquad \mathcal{V}_T = \{\nu^{(0)}\} \cup \{\nu^{(i)} : i \in \mathcal{I}_T\}.$$

Assume $H_{\min} \to \infty$ and fixed positive constants satisfy $\Delta_i \geq \underline{\Delta}$, $\eta_i \geq \underline{\eta}$, and $0 < d_i \leq \overline{d}$ uniformly over the indicated arms and all sufficiently large T. Let a policy sequence obey

$$g_T := \sup_{\nu \in \mathcal{V}_T} \mathbb{E}_\nu R_T(\pi^{(T)}; \nu) = o(H_{\min}^a) \quad \text{for every } a > 0.$$

Proof of the local-alternative regret lower bound. Fix $i \in \mathcal{I}_T$ and let

$$A_i = \{N_i(T) \leq H_i/2\}.$$

Under the baseline, every pull of i incurs regret Δ_i, so $\mathbb{E}_0 N_i(T) \leq g_T/\underline{\Delta}$. Markov's inequality gives

$$\mathbb{P}_0(A_i^c) \leq \frac{2g_T}{\underline{\Delta} H_i}. \tag{17}$$

Under $\nu^{(i)}$, arm i is uniquely best after its arrival. Each post-arrival pull of a different arm incurs regret at least η_i. On A_i there are at least $H_i/2$ such pulls. Therefore

$$\mathbb{P}_{\nu^{(i)}}(A_i) \leq \frac{2g_T}{\underline{\eta} H_i}. \tag{18}$$

Take $a = 1/2$ in the uniform-goodness condition. For all sufficiently large T, $g_T \leq H_{\min}^{1/2}$. Consequently the right side of (17) is at most $2/(\underline{\Delta} H_{\min}^{1/2})$, which tends to zero uniformly in i. The right side of (18) is at most $2/(\underline{\eta} H_i^{1/2})$, which is at most $H_i^{-1/3}$ once $H_{\min}$ is sufficiently large. Hence one common deterministic threshold in T gives, for every $i \in \mathcal{I}_T$,

$$\mathbb{P}_0(A_i) \geq \frac{1}{2}, \qquad \mathbb{P}_{\nu^{(i)}}(A_i) \leq H_i^{-1/3}.$$

Applying (13) and (16) yields

$$d_i \mathbb{E}_0 N_i(T) \geq \frac{1}{6} \log H_i - \log 2 \geq \frac{1}{12} \log H_i$$

for all sufficiently large T, uniformly over i. In the last step use $H_i \geq H_{\min} \to \infty$. If an alternative event had probability zero, the binary divergence would be infinite while the adaptive divergence is finite; the displayed argument is valid with the usual extended-value convention as well.

Under the same baseline environment for all the arms,

$$R_T(\pi^{(T)}; \nu^{(0)}) \geq \sum_{i \in \mathcal{I}_T} \Delta_i N_i(T).$$

There is no summation of lower bounds obtained in incompatible baseline environments. Taking expectations and summing the uniform armwise bounds proves

$$\mathbb{E}_0 R_T \geq \frac{1}{12} \sum_{i \in \mathcal{I}_T} \frac{\Delta_i}{d_i} \log H_i \geq \frac{\underline{\Delta}}{12\overline{d}} |\mathcal{I}_T| \log H_{\min}.$$

□

For a static fixed collection of K inferior arms, all $H_i = T$. Fixed positive gaps and alternative improvements give positive lower bounds $\underline{\Delta}, \underline{\eta}$, and finitely many finite divergences have a finite common upper bound. If additionally $d_i \le C_i \Delta_i^2$, the preceding display becomes

$$\mathbb{E}_0 R_T \ge \frac{1}{12} \sum_{i=1}^{K} \frac{\log T}{C_i \Delta_i}.$$

This proves the static specialization directly. Its fixed laws and fixed K distinguish it from a growing triangular family. The uniform-goodness hypothesis is a condition on policies; the result does not establish that this policy class is nonempty for every growth schedule. In particular, its two conclusions together imply $|\mathcal{I}_T| \log H_{\min} = o(H_{\min}^a)$ for every $a > 0$ whenever the assumed sequence exists.

S2.3 Successive local comparisons under benchmark improvements

Fix $J \ge 1$, positive integer block lengths H_m with $\sum_{m=1}^{J} H_m = T$, and endpoints $s_m = \sum_{\ell=1}^{m} H_\ell$, $s_0 = 0$. At time $s_{m-1} + 1$, two Bernoulli arms c_m, d_m arrive, with means

$$r_{c_m} = \theta_m, \qquad r_{d_m} = \theta_m + \sigma_m \varepsilon_m, \qquad \sigma \in \{-1, +1\}^J.$$

The arms remain available. There are no other arms. Fix $0 < \gamma < 1/2$, assume $\varepsilon_m > 0$ and $\theta_m \pm \varepsilon_m \in [\gamma, 1 - \gamma]$, and require $\theta_{m+1} \ge \theta_m + \varepsilon_m$ for $m < J$. Let ν_σ denote the corresponding full environment.

Proof of the deterministic-block regret lower bound. For a fixed policy, let

$$\mathcal{R}_m = \sum_{t=s_{m-1}+1}^{s_m} (r^*(t) - r_{I_t}), \qquad N_m = \sum_{t=s_{m-1}+1}^{s_m} \mathbf{1}\{I_t = d_m\}.$$

These are random variables, and $R_T = \sum_m \mathcal{R}_m$ pathwise. Fix a block m and fix all signs other than σ_m. Denote the two resulting environments by $\nu^{m,-}$ and $\nu^{m,+}$. For every earlier block $j < m$,

$$r_{c_j} = \theta_j \le \theta_m, \qquad r_{d_j} \le \theta_j + \varepsilon_j \le \theta_{j+1} \le \theta_m.$$

Thus the best mean in block m is θ_m under the negative sign and $\theta_m + \varepsilon_m$ under the positive sign. All arms older than block m remain available; they cannot exceed these reference levels. Consequently,

$$\mathcal{R}_m \ge \varepsilon_m N_m \quad \text{under } \nu^{m,-}, \qquad \mathcal{R}_m \ge \varepsilon_m (H_m - N_m) \quad \text{under } \nu^{m,+}.$$

Let $P_{[m]}^{m,-}$ and $P_{[m]}^{m,+}$ be the history laws only up to time s_m. The variable N_m / H_m is measurable

for that prefix and lies in $[0,1]$. By (14),

$$\begin{aligned}\mathbb{E}_{m,-}\mathcal{R}_m + \mathbb{E}_{m,+}\mathcal{R}_m &\geq \varepsilon_m\{\mathbb{E}_{m,-}N_m + \mathbb{E}_{m,+}(H_m - N_m)\}\\ &\geq H_m\varepsilon_m\{1 - \mathrm{TV}(P_{[m]}^{m,-}, P_{[m]}^{m,+})\}.\end{aligned}$$

Only the observations of d_m have different laws. That arm has no observations before its arrival at the beginning of block m. Equation (16), applied at the prefix endpoint s_m, therefore gives

$$D(P_{[m]}^{m,-}\|P_{[m]}^{m,+}) = \mathbb{E}_{m,-}N_m\ \mathrm{kl}(\theta_m - \varepsilon_m, \theta_m + \varepsilon_m) \leq H_m\,\mathrm{kl}(\theta_m - \varepsilon_m, \theta_m + \varepsilon_m).$$

Using a full-horizon transcript instead would include later pulls of d_m; the prefix endpoint is essential to the bound by H_m. The Bernoulli upper bound and Pinsker's inequality give

$$\mathrm{TV}(P_{[m]}^{m,-}, P_{[m]}^{m,+}) \leq \sqrt{\frac{1}{2}H_m\,\mathrm{kl}(\theta_m - \varepsilon_m, \theta_m + \varepsilon_m)} \leq \sqrt{\frac{2H_m\varepsilon_m^2}{\gamma(1-\gamma)}}.$$

The pair of expected block regrets is nonnegative, so it is at least

$$H_m\varepsilon_m\left(1 - \sqrt{\frac{2H_m\varepsilon_m^2}{\gamma(1-\gamma)}}\right)_+.$$

Averaging over the two signs in this block contributes a factor $1/2$. Averaging also over all fixed configurations of the other signs, and then summing over blocks, gives

$$\begin{aligned}\sup_\sigma \mathbb{E}_{\nu_\sigma}R_T &\geq 2^{-J}\sum_{\sigma\in\{-1,+1\}^J}\mathbb{E}_{\nu_\sigma}R_T\\ &\geq \frac{1}{2}\sum_{m=1}^{J}H_m\varepsilon_m\left(1 - \sqrt{\frac{2H_m\varepsilon_m^2}{\gamma(1-\gamma)}}\right)_+.\end{aligned}$$

The same uniform sign average is used for every block, so these blockwise bounds combine under one full-environment distribution. If $H_m\varepsilon_m^2 \leq \kappa\gamma(1-\gamma)$ with $0 < \kappa < 1/2$, each factor is at least $1 - \sqrt{2\kappa}$. The proof holds for every policy, including randomized and horizon-dependent policies, and therefore also after taking their infimum. □

For completeness, if $\varepsilon_m \geq c_\varepsilon\sqrt{J/T}$ and the small-information condition holds, the last bound is at least

$$c_\kappa\sum_{m=1}^{J}H_m\varepsilon_m \geq c_\kappa c_\varepsilon\sqrt{JT}.$$

Conditions $H_m \asymp T/J$ and $\varepsilon_m \asymp \sqrt{J/T}$ with a sufficiently small product of upper constants ensure that small-information condition. The ladder must also exist inside the bounded interval.

Telescoping the monotonicity constraints gives

$$\sum_{m=1}^{J} \varepsilon_m \le \theta_J + \varepsilon_J - (\theta_1 - \varepsilon_1) \le 1 - 2\gamma.$$

Consequently $c_\varepsilon J^{3/2}/\sqrt{T} \le 1 - 2\gamma$, proving the necessary range $J = O(T^{1/3})$. A homogeneous example with $\varepsilon_m = \varepsilon$ and $\theta_m = \gamma + m\varepsilon$ is feasible whenever $(J+1)\varepsilon \le 1 - 2\gamma$. There are exactly $2J$ arrived arms, so block count and arm count are different quantities. These deterministic-horizon lower bounds do not establish matching with an upper bound at a random stopping time, nor do they impose a growth boundary on all possible arriving-arm environments.

S3 Probability tools and reporting conventions

For the reporting constructions, index arms in arrival order $i = 1, 2, \ldots$, with a fixed order within each batch. Every arm has an independent potential reward stream $(Y_{i,n})_{n\ge1}$ in $[0,1]$. The policy observes its nth entry on its nth pull. Arrival times are deterministic, and reports are issued after the current observation. Write $N_i(t)$ for lifetime observations through round t and $\hat{r}_i(n) = n^{-1}\sum_{s\le n} Y_{i,s}$. The observed reward in the original physical-time notation is $X_{t,I_t} = Y_{I_t,N_{I_t}(t-1)+1}$. The conclusions below also hold conditionally on an independent exogenous arrival schedule.

For independent $[0,1]$ observations, Hoeffding's inequality gives

$$P\{|\hat{r}_i(n) - r_i| > x\} \le 2e^{-2nx^2}.$$

For completeness, the centered log moment generating function has value and derivative zero at the origin; its second derivative is a variance under an exponentially tilted distribution and is at most $1/4$. Integrating twice gives $Ee^{\lambda(X-EX)} \le e^{\lambda^2/8}$. Multiplication over independent observations and exponential Markov inequality prove the display.

With

$$b_i(n) = \sqrt{\frac{\log(8i^2n^2/\delta)}{2n}},$$

the union of all potential-prefix failures has probability at most $\delta\zeta(2)^2/4 < \delta$. Intersecting the prefix intervals with $[0,1]$ therefore gives simultaneous coverage. On that event each endpoint differs from its mean by at most $2b_i(n)$. No conditioning on a previously selected empirical mean is used. Empty intersections on failure paths are handled by exclusion-first reporting; the reward policy remains defined on all paths.

The checkpoint alternative fixes, before observing arm i,

$$q_i(\rho_i) = \left\lceil \frac{2}{\rho_i^2} \log \frac{4i(i+1)}{\delta} \right\rceil.$$

Before q_i observations its interval is $[0,1]$; afterwards it is permanently

$$[\hat{r}_i(q_i) - \rho_i/2, \hat{r}_i(q_i) + \rho_i/2] \cap [0,1].$$

The sum of coverage failures is at most $\sum_i \delta/\{2i(i+1)\} = \delta/2$. On coverage, each endpoint differs from the true mean by at most ρ_i. These intervals are nonempty on every path and only contract once. If $\underline{r}^*(t) = \max_i \ell_i(t)$ and $\overline{r}^*(t) = \max_i u_i(t)$, use

$$\mathcal{P}_t = \{i : \ell_i(t) \geq \overline{r}^*(t) - \varepsilon\}, \quad \mathcal{N}_t = \{i : u_i(t) < \underline{r}^*(t) - \varepsilon\}, \quad \mathcal{V}_t = \mathcal{A}_t \setminus (\mathcal{P}_t \cup \mathcal{N}_t).$$

The sets $\mathcal{P}_t, \mathcal{N}_t$ are disjoint because $\underline{r}^*(t) \leq \overline{r}^*(t)$ and $\ell_i \leq u_i$. Coverage implies $\mathcal{P}_t \subseteq \mathcal{G}_t^\varepsilon \subseteq \mathcal{A}_t \setminus \mathcal{N}_t$. Moreover, $\mathcal{N}_t$ is increasing as observations and arrivals accumulate: $\underline{r}^*(t)$ cannot decrease and an existing u_i cannot increase.

For two Bernoulli mean vectors $\boldsymbol{r} = (r_i)_i$ and $\boldsymbol{r}' = (r'_i)_i$ and a bounded stopping time τ, action kernels, policy randomization, and stopping kernels cancel in the history likelihood ratio. Padding after stopping by null symbols and taking iterated expectations yields

$$D(P_{\boldsymbol{r}}^{H_\tau} \| P_{\boldsymbol{r}'}^{H_\tau}) = \sum_i E_{\boldsymbol{r}} N_i(\tau) \, \mathrm{kl}(r_i, r'_i). \tag{19}$$

The same proof applies to arbitrary observation laws with finite two-way KL. For an event F measurable at stopping, splitting the likelihood integral over F, F^c and applying Jensen gives

$$D(P\|Q) \geq \mathrm{kl}(P(F), Q(F)).$$

We repeatedly use

$$2(p-q)^2 \leq \mathrm{kl}(p,q) \leq \frac{(p-q)^2}{q(1-q)}.$$

The upper bound follows from $\log x \leq x - 1$; for the lower bound the second derivative in p is $1/\{p(1-p)\} \geq 4$ and both value and derivative vanish at $p = q$.

S4 The broadcast obstruction and its queue consequence

Precise claim. Let $0 < \varepsilon \leq 1/16$, $0 < \delta \leq 1/16$, $0 < \eta < \varepsilon/2$. The reference arm has mean $1/4 + \varepsilon/2$. All candidates have mean $p = 1/4$ in environment zero; environment j raises only candidate j to $p + 2\varepsilon$. Let M_t be the candidate count, a_t the new-candidate count, and

$$q_- = \left\lfloor \frac{\mathrm{kl}(1/4, \delta)}{\mathrm{kl}(1/4, 1/4 + 2\varepsilon)} \right\rfloor, \quad Q_t = (Q_{t-1} + q_- a_t - 1)_+, \quad Q_0 = 0,$$

$$\mathcal{B}_T = \sum_{t \leq T} (M_t - 1)_+ 1\{Q_t > 0\}.$$

Every policy reliable on these environments obeys

$$E_0\mathcal{U}_T(\eta) \geq (3/4 - \delta)\mathcal{B}_T, \qquad P_0\{\mathcal{U}_T(\eta) \geq \mathcal{B}_T/4\} \geq 1/4$$

when $\mathcal{B}_T > 0$.

Proof. The denominator defining q_- is at most $\mathrm{kl}(1/4, 3/8) \leq 1/15$, whereas its numerator is at least $\frac{1}{4}\log 4 + \frac{3}{4}\log(4/5) \geq 7/48 > 2/15$. Thus $q_- \geq 2$. The recursion gives, by induction,

$$Q_t = \max_{0 \leq u \leq t} \left\{ q_- \sum_{r=u+1}^{t} a_r - (t-u) \right\}.$$

If $Q_t > 0$, there is a suffix in which total candidate demand exceeds all physical observation slots. Its arms had no observations before that suffix. Consequently at least one deterministic candidate label j satisfies $E_0 N_j(t) < q_-$. The choice is made from expected counts, not from a realized history. For any other candidate i, membership of i in $\mathcal{P}_t$ is erroneous under environment j. Its probability there is at most δ. The history KL is strictly below $\mathrm{kl}(1/4, \delta)$, so data processing and monotonicity of binary KL force $P_0(i \in \mathcal{P}_t) < 1/4$. Since all candidates are genuinely good under environment zero, $P_0(i \in \mathcal{N}_t) \leq \delta$. It follows that $P_0(i \in \mathcal{V}_t) \geq 3/4 - \delta$. These candidates all have boundary distance $\varepsilon/2 > \eta$. Summing at deterministic busy rounds proves the expectation bound, entirely under the one environment zero.

Let Y count candidate unresolved area only at busy rounds with $M_t \geq 2$. Then $Y \leq \mathcal{U}_T$, $0 \leq Y \leq 2\mathcal{B}_T$, and $E_0 Y \geq 11\mathcal{B}_T/16$. If $p = P_0(Y \geq \mathcal{B}_T/4)$, then $11\mathcal{B}_T/16 \leq \mathcal{B}_T/4 + (7\mathcal{B}_T/4)p$, so $p \geq 1/4$. □

Queue area. For an integer $q \geq 2$, suppose m candidates arrive by time s and put $D = s + qm - 1$. Then

$$\sum_{t \leq D} (M_t - 1)_+ 1\{Q_t(q) > 0\} \geq \frac{q-1}{2} m(m-1).$$

To prove this, remove all subsequent arrivals, which can only decrease both the queue and its weights. The qm service units finish by D. At the nth actual service, at least $\lceil n/q \rceil$ candidates have arrived, so service rounds weighted by $M_t - 1$ sum to at least $\sum_{n=1}^{qm} (\lceil n/q \rceil - 1) = qm(m-1)/2$. The post-service queue is zero at the last service round of each busy period. Their cumulative arrival counts are distinct integers in $\{1, \ldots, m\}$, so removing their weights costs at most $m(m-1)/2$. This proves the claim.

In particular, if along a fixed infinite schedule a policy reliable in every environment has a deterministic area envelope $a_T = o(T)$, then $K_t = o(\sqrt{t})$. Otherwise along s_n there are at least $c\sqrt{s_n}$ candidates. Choose the first $m_n = \lfloor c\sqrt{s_n} \rfloor$; eventually $q_- m_n \leq s_n$. The queue bound and the preceding probability statement at $2s_n$ give $P_0(\mathcal{U}_{2s_n} \geq c' s_n) \geq 1/4$, contradicting $\delta \leq 1/16$ and the proposed envelope.

S5 The partial simultaneous-recovery lemma

Precise claim. Let $m \geq 1$ independent uniform signs specify independent observation laws P_i^+, P_i^-. All nuisance parameters and observations are independent of these signs. Sampling may be adaptive and interleaved, and τ is bounded. At every sign configuration, the probability of any erroneous determinate report is at most $\delta \leq 1/16$. Let J_τ be the number of determinate reports and $\kappa = \max_i \max\{D(P_i^+ \| P_i^-), D(P_i^- \| P_i^+)\} \in (0, \infty)$. If $\overline{P}(J_\tau \geq m/2) \geq 1/2$, then

$$\overline{E} \sum_i N_i(\tau) \geq \frac{m}{1024\kappa} \log \frac{m}{\delta}.$$

Proof. Let L_i be the observed positive-versus-negative log likelihood ratio for coordinate i. After conditioning on the full stopped history, the action and stopping kernels cancel. The likelihood factors over coordinates and so does the posterior, even though the sample counts need not be independent. The smallest coordinatewise posterior error is $e_i = (1 + e^{|L_i|})^{-1}$. If S is the reported subset, its conditional probability of complete correctness is at most $\prod_{i \in S}(1 - e_i)$; choosing a non-MAP label cannot improve it. Thus

$$\overline{E}\left[1 - \prod_{i \in S}(1 - e_i)\right] \leq \delta.$$

With probability at least $1 - 2\delta$ this product is at least $1/2$ and hence $\sum_{i \in S} e_i \leq \log 2$. Intersecting with $J_\tau \geq m/2$ costs at most 2δ and leaves probability at least $3/8$. For $m \geq 4096$, Jensen gives on this intersection

$$\sum_i |L_i| \geq \sum_{i \in S} \{\log(1/e_i) - \log 2\} \geq \frac{m}{2} \log \frac{m}{4 \log 2}.$$

Successive conditional expectations of signed likelihood increments give

$$\overline{E} \sum_i \sigma_i L_i \leq \kappa \overline{E} \sum_i N_i(\tau).$$

Finite two-way KL makes the stopped increments integrable. Given the history, $E(\sigma_i L_i \mid H_\tau) = |L_i|(1 - 2e_i)$, and $2|L_i|e_i \leq 1$ for every real L_i. Therefore

$$\kappa \overline{E} \sum_i N_i \geq \overline{E} \sum_i |L_i| - m \geq m \left\{ \frac{3}{16} \log \frac{m}{4 \log 2} - 1 \right\} \geq \frac{m}{256} \log m,$$

where the last inequality holds for $m \geq 4096$.

For the confidence term, fix a sign vector and flip coordinate i only. The event reporting the original sign on coordinate i has original probability at least its determinate-report probability minus δ, and flipped probability at most δ. Average the data-processing inequalities over vectors and coordinates and use joint convexity of binary KL. Since $\overline{E} J_\tau \geq m/4$, this yields

$$\kappa \overline{E} \sum_i N_i \geq m \, \mathrm{kl}(1/4 - \delta, \delta) \geq \frac{m}{256} \log(1/\delta).$$

To check the last scalar inequality, write $L = \log(1/\delta)$. For $L \le 8$, Pinsker gives a lower bound $2(1/4 - 2\delta)^2 \ge 1/32 \ge L/256$. For $L \ge 8$, binary entropy at most $\log 2$ gives $\mathrm{kl}(1/4 - \delta, \delta) \ge 3L/16 - \log 2 \ge L/16$. When $m \ge 4096$, the maximum of the two preceding bounds is at least half their sum. When $m < 4096$, $\log m \le 3\log(1/\delta)$, and the confidence bound alone suffices. Both cases imply the stated constant $1/1024$. □

Single-window truncation. If such a batch has H rounds with qualifying boundary distance larger than η, set

$$h_* = \min\left\{H, \left\lceil \frac{m\log(m/\delta)}{1024\kappa} \right\rceil - 1\right\}.$$

At every $r \le h_*$ there are at most r batch observations. The recovery lemma's contrapositive makes the probability of half-completion less than $1/2$, so the expected unresolved count is at least $m/4$. Thus batch area has expectation at least $mh_*/4$. Its first-h_*-round area Y lies in $[0, mh_*]$; consequently $\overline{P}(Y \ge mh_*/8) \ge 1/7$. One fixed sign configuration attains at least this probability. If $h_* = 0$, only the trivial zero lower bound is asserted.

S6 Area-constrained multiple updates

In window e, suppose M_e candidates have gaps $\varepsilon - \sigma_{e,k}d_e$, and a new reference makes their signs irrelevant after that window. Put $w_e = M_eH_e$, $V = \sum_e w_e$, and let p_e be the product-prior probability that at least half of batch e becomes determinate simultaneously before its window ends. Define

$$\Phi_\delta(u) = \begin{cases} \mathrm{kl}(1 - \delta - u, \delta), & u < 1 - 2\delta, \\ 0, & u \ge 1 - 2\delta, \end{cases}$$

$$g_{m,\delta}(p) = \max\left\{0, \frac{(p - 2\delta)_+}{2}\left[\log\frac{m}{4\log 2}\right]_+ - 1, \Phi_\delta(1 - p/2)\right\}.$$

Each displayed component is convex and nondecreasing in p.

Proof of the general area-to-regret lower bound. If window e never half-completes, at every round more than half its candidates remain unresolved. Its area is at least $w_e/2$. On area success, $\sum_e w_e 1_{S_e^c} \le 2a$; outside that event this sum is at most V. Consequently

$$\sum_e w_e p_e \ge (1 - \delta)V - 2a.$$

Stop batch e at first half-completion or at its endpoint. The likelihood argument above now intersects the product-posterior event with completion probability p_e, giving intersection probability at least $(p_e - 2\delta)_+$ and expected determinate count at least $M_e p_e/2$. The absolute-likelihood and coordinate-flip derivations therefore give

$$\kappa_e \overline{E} N_e \ge M_e g_{M_e,\delta}(p_e).$$

For the absolute-likelihood term, when the bracketed logarithm is positive its Jensen expression is increasing for subset sizes at least $M_e/2$; when it is nonpositive the stated term is at most zero. Each observation counted before stopping has regret at least $\varepsilon - d_e$. The signs of all batches belong to one product prior, and samples of different batches are disjoint. Hence

$$\sup_{\sigma} E_\sigma R_T \geq \inf_{\substack{0\leq p_e\leq 1 \\ \sum_e w_e p_e \geq (1-\delta)V - 2a}} \sum_e \frac{(\varepsilon - d_e)M_e}{\kappa_e} g_{M_e,\delta}(p_e).$$

This is an information lower relaxation, not a claim that every feasible allocation is achieved by a causal policy. □

Homogeneous consequence. Take $M_e = M$, $H_e = H$, $d_e = d$, and $a = JMH/8$. On success at least $3J/4$ windows half-complete; thus $\sum_e p_e \geq 45J/64$. At least $3J/8$ windows have $p_e \geq 1/2$: otherwise the sum is at most $3J/8 + (5J/8)/2 = 11J/16 < 45J/64$. The Bernoulli means lie in $[1/4, 3/4]$, giving $\kappa_e \leq 32d^2$. The partial-recovery lemma and $\varepsilon - d \geq \varepsilon/2$ yield

$$\sup_{\sigma} E_\sigma R_T \geq \frac{3J}{8} \frac{\varepsilon M}{2 \cdot 1024 \cdot 32 d^2} \log \frac{M}{\delta} \geq 2^{-20} \frac{\varepsilon JM}{d^2} \log \frac{M}{\delta}.$$

S7 History-retaining checkpoint UCB and evidence reuse

CP-UCB-AA fixes each arm's checkpoint before seeing its data. At even physical rounds it samples the earliest-arriving arm below its lifetime checkpoint, if one exists. At all other rounds it maximizes, over every available arm including reported exclusions,

$$J_i(t) = \hat{r}_i(N_i(t-1)) + \sqrt{3\log(2t)/N_i(t-1)},$$

with index $+\infty$ at zero count and deterministic ties. Every observation serves both the lifetime checkpoint and the empirical reward estimate.

A reward bound that does not condition on report coverage. At physical time t, order arms by their first observed rewards. At most t streams have been exposed, each at at most t potential prefixes. A previously unobserved stream is independent of the history selecting it. Applying Hoeffding to each newly exposed stream and union bounding gives

$$P\left\{\exists \text{ observed arm and prefix } n \leq t : |\hat{r}_i(n) - r_i| > \sqrt{3\log(2t)/n}\right\} \leq 2t^2(2t)^{-6} = \frac{1}{32t^4}.$$

The exposure order is used only for this union bound, not to assert independence among observed counts. An ordinary UCB pull with finite index, on this event's complement, has $\Delta_{I_t}(t) \leq 2\sqrt{3\log(2t)/N_{I_t}(t-1)}$. Each arm–pre-pull-count pair is charged at most once; interleaved service

observations skip counts. Therefore the ordinary, nonfirst-pull regret is at most

$$4\sqrt{3\log(2T)}\sum_i \sqrt{N_i(T)} \le 4\sqrt{3K_T T\log(2T)}$$

on good rounds. First pulls cost at most K_T, and the sum of expected bad-round losses is less than one. Adding the actual regret of service rounds proves

$$ER_T \le \min\left\{T, K_T + 4\sqrt{3K_T T\log(2T)} + 1 + E\sum_{t\in\mathcal{I}_T}\Delta_{I_t}(t)\right\}.$$

There is no δT term: the reward policy never permanently discards an arm because of a reporting error.

Arbitrary-arrival completion bound. At any fixed reporting precision $\rho = \eta/4$, if all current checkpoints are complete, all endpoint errors are at most $\eta/4$ and every arm whose boundary distance exceeds η has a determinate report. A virtual FIFO queue adding q_i work at each arrival and serving one unit at every even round dominates unfinished checkpoints; additional UCB observations only reduce actual work. Its busy periods, including alignment to the first even round, have total length at most $2\sum_i q_i$. Thus on simultaneous coverage,

$$\mathcal{U}_T(\eta) \le \sum_{t\le T} K_t 1\{\overline{Q}_t > 0\} \le \min\{K_T T, 2K_T\sum_{i\le K_T} q_i\}.$$

At fixed η, δ, the checkpoints are $O(\log(ei))$, so $K_T^2\log(eK_T) = o(T)$ suffices for both sublinear area and expected regret. Combined with the broadcast consequence, a path with $K_t \asymp t^\alpha$, $0 < \alpha < 1$, admits both if and only if $\alpha < 1/2$. This is an exponent statement and does not remove the critical logarithm.

Separated windows and reuse. Let batch e have reference mean b_e, M_e candidate means $b_e - \varepsilon \pm d_e$, and common predetermined precision ρ_e. The final batch contains only its reference; set $M_J = M_{-1} = 0$ and $S_e = \sum_{i\in F_e} q_i(\rho_e)$. Assume

$$H_e \ge 2S_e, \quad b_J - b_0 \le \varepsilon/4, \quad d_e + \rho_e + \rho_{e+1} < b_{e+1} - b_e, \quad 2\max_e \rho_e \le \varepsilon/2.$$

The algorithm is not told which arm is the reference or when the next window starts. By induction every batch completes its checkpoints within its first $2S_e$ rounds, before the next batch appears. On coverage, successive reference levels satisfy $b_e + \rho_e < b_{e+1} - \rho_{e+1}$. Once batch e completes, the maximum upper endpoint of all available arms is at most $b_e + \rho_e$. Its candidates are therefore completely classified when $2\rho_e < d_e$.

An old candidate in batch $e-1$ has upper endpoint at most $b_{e-1} - \varepsilon + d_{e-1} + \rho_{e-1}$, whereas the new reference has lower endpoint at least $b_e - \rho_e$. Their difference exceeds ε, excluding the entire preceding candidate batch without any further old-arm observations. Older exclusions persist.

Every reference has gap at most $\varepsilon/4$ and total endpoint error at most $\varepsilon/2$, so all references become positively certified. If $F = \{e < J : 2\rho_e < d_e\}$, the area is consequently at most

$$\sum_{e \notin F} M_e H_e + 2 \sum_{e \in F} M_e S_e + 2 \sum_{e=0}^{J} (e + 1 + M_{e-1}) S_e.$$

The terms cover current nonfine candidates, fine candidates, and temporarily unresolved references plus the preceding candidate batch. All earlier candidate batches have already been excluded.

For expected regret, every observation with pre-pull count below q_i occurs in its arm's birth window. Such reference observations have zero regret, and candidate observations cost at most $\varepsilon + d_e$ each. All observations with pre-pull count at least q_i are ordinary UCB pulls. Let γ_i be the smallest positive gap arm i ever has and D_i^T its largest gap through T. At a good UCB round, any positive-gap pull has pre-pull count at most $12 \log(2T)/\gamma_i^2$. Since counts do not repeat, this gives the unconditional bound

$$ER_T \le \sum_{e<J} \sum_{k=1}^{M_e} (\varepsilon + d_e) q_{e,k} + \sum_{i:\gamma_i \text{ exists}} D_i^T \left[\left\lceil \frac{12 \log(2T)}{\gamma_i^2} \right\rceil - q_i + 1 \right]_+ + 1.$$

The completion-before-next-batch assumption is essential to the first charging term. It is not an assertion about arbitrary overlapping arrivals.

S8 The rapid and deferred endpoints on one family

Here $J \ge 1$, $M \ge \max\{2, J\}$, $0 < \varepsilon \le 1/8$, $0 < \delta \le 1/16$, $Jv \le \varepsilon/4$, $0 < d \le v/32$, and $\eta = d/2$. The reference means are $b_e = 1/2 + ev$ and the M candidates born in window $e < J$ have means $b_e - \varepsilon + \sigma_{e,k} d$. Each window has length H, and the final window contains only a new reference. Set

$$K = J(M+1) + 1, \quad L_K = \log \frac{4K(K+1)}{\delta}, \quad q_f = \lceil 32 L_K / d^2 \rceil, \quad q_c = \lceil 128 L_K / v^2 \rceil,$$

$$H = 128(M + J + 1) q_f, \qquad T = (J+1)H, \qquad V = JMH.$$

These are analysis quantities: the policy uses the selected precision $\rho_f = d/4$ or $\rho_c = v/8$, not K, H, T or the best-arm identities. Both choices satisfy the separated-window conditions.

The preceding area bound is at most $B_{J,M} q_f$ for the fine policy and $V + B_{J,M} q_c$ for the coarse policy, where

$$B_{J,M} = 2(M+1)\{J(2M+1) + J(J-1)/2\} + 2(M + J + 1).$$

For $2 \le M$ and $J \le M$, $B_{J,M} \le 12JM^2$. Indeed, $M + 1 \le 3M/2$, $2M + 1 \le 5M/2$, and $J - 1 \le M$ bound the main term by $9JM^2$, and the final term by $3JM^2$. The definition of H gives $B_{J,M} q_f \le V/8$, while $q_c \le q_f$ gives coarse area at most $9V/8$. Both statements hold on the same fixed-checkpoint coverage event of probability at least $1 - \delta/2$.

For rapid maintenance the area-to-regret lower bound applies. For its upper bound, every available arm in this family has gap at most 2ε. Hence, on every path, $R_T \le 2\varepsilon T$. Since

$(J+1)(M+J+1)/(JM) \le 5$,

$$2\varepsilon T \le 1280\varepsilon JMq_f.$$

Furthermore $K \le 2M^2$, $K+1 \le 3M^2$ imply $L_K \le 5\log(M/\delta)$, and $q_f \le 165d^{-2}\log(M/\delta)$. Thus

$$2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta} \le \mathfrak{R}_\Theta(V/8) \le 2^{18}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}.$$

This upper bound uses the stated horizon and must not be extended unchanged to arbitrarily long windows.

For the coarse policy, within-checkpoint candidate regret is at most $2\varepsilon JMq_c$. Candidates have $\gamma_i \ge \varepsilon - d \ge 3\varepsilon/4$, $D_i^T \le 2\varepsilon$; old references have $\gamma_i \ge v$, $D_i^T \le Jv$. Dropping the favorable quota subtraction in the preceding reward bound and absorbing integer terms gives

$$\sup_\Theta ER_T \le 2\varepsilon JMq_c + 64\left(\frac{JM}{\varepsilon} + \frac{J^2}{v}\right)\log(2T) + 1.$$

At fixed $J, M, \varepsilon, v, \delta$, $d \downarrow 0$ makes $T = \Theta(d^{-2})$. Consequently the rapid endpoint is $\Theta(d^{-2})$ and the deferred endpoint is at most $O(\log(1/d))$. No matching lower bound for the latter endpoint is asserted.

S9 Eventual exact classification and unconditional regret

Let epochs be $E_e = \{16^e, \dots, 16^{e+1} - 1\}$ and use cumulative targets $q_e = \lceil 4^e(e+1)^2 \rceil$. At even rounds serve, by FIFO, any arrived arm below the current target, including arms formerly reported excluded. Other rounds use full-history UCB. In epoch e report from the raw intervals with radius

$$b_{i,e}(n) = \sqrt{\frac{\log\{16i^2(e+1)^2n^2/\delta\}}{2n}},$$

clipped to $[0,1]$ and equal to $[0,1]$ at zero count. Intervals are recomputed rather than permanently intersected across epochs.

Hoeffding and a union over i, e, n bound the probability of any coverage failure by $\delta\zeta(2)^3/8 < \delta$. This proves all-time reporting reliability. Suppose arrivals eventually stop with $K < \infty$. For all large e, the current epoch's remaining checkpoint work is at most Kq_e, and its $2Kq_e$ physical service time is smaller than $15 \cdot 16^e$. Therefore each arm has at least q_{e-1} observations at the beginning of every sufficiently late epoch. Uniformly for $t \in E_e$ and actual $n \in [q_{e-1}, t]$, the radius is $O(\sqrt{\{e + \log i + \log(1/\delta)\}/q_{e-1}}) = o(1)$.

For every positive rational z, Hoeffding makes $\sum_n P(|\hat{r}_i(n) - r_i| > z) < \infty$. Tail union bounds imply only finitely many deviations of size z almost surely. Taking a countable intersection over z and the finitely many final arms gives simultaneous empirical-mean convergence. If $\min_i |\max_j r_j - r_i - \varepsilon| > 0$, then all interval comparisons are eventually correct and complete, almost surely. This conclusion is not conditioned on early coverage, and early reporting errors can be corrected.

Through $T \in E_e$, service pulls number at most Kq_e, because targets are cumulative and lifetime counts never decrease. This is $O(K\sqrt{T}(1+\log T)^2)$. Apply the unconditional UCB counting bound to the remaining pulls to obtain

$$ER_T = O\{K\sqrt{T}(1+\log T)^2 + \sqrt{KT\log(2T)}\} = o(T)$$

for each eventually finite instance. Exact boundary cases are not included in the complete-classification claim.

S10 Worst-schedule area and a sparse but bursty schedule

Suppose $0 < \varepsilon \le 1/8$, $0 < \eta \le \varepsilon/32$, $0 < \delta \le 1/16$, and $K \ge 2$. Let $\mathfrak{A}_{T,K}$ be the smallest area threshold attainable with joint label-and-area probability at least $1-\delta$, uniformly over reward environments and schedules with $K_T \le K$. The policy must be reliable on the full model.

Upper bound. Use fixed precision $\rho = \eta/4$. Then

$$q_i = \left\lceil 32\eta^{-2}\log\frac{4i(i+1)}{\delta}\right\rceil \le 99\eta^{-2}\log(K/\delta), \qquad i \le K.$$

The arbitrary-arrival queue bound proves, on coverage,

$$\mathcal{U}_T \le \min\{KT, 2K\sum_{i\le K} q_i\} \le 256\min\{KT, K^2\eta^{-2}\log(K/\delta)\}.$$

The UCB reward bound and at most $\sum_i q_i$ service pulls also give

$$ER_T \le \min\{T, K + 4\sqrt{3KT\log(2T)} + 1 + 99K\eta^{-2}\log(K/\delta)\}.$$

Lower bound. At round one release a reference of mean $1/2+\varepsilon$ and $m = K-1$ candidates of means $1/2 \pm 2\eta$. Their signs are independent and specify their relative-goodness labels. The reference is always optimal; its known distribution gives no sign information. Boundary distances equal $2\eta > \eta$, and two-way KL is at most $128\eta^2$. Set

$$q_* = \frac{m\log(m/\delta)}{131072\eta^2}, \qquad h = \min\{T, \lceil q_*\rceil - 1\}.$$

Since $\eta \le 1/256$, $q_* \ge m\log(m/\delta)/2 > 1$ and $h \ge \frac{1}{2}\min\{T, q_*\}$. Single-window truncation gives a fixed environment with

$$P\{\mathcal{U}_T \ge mh/8\} \ge 1/7 > \delta.$$

Therefore no valid uniform area quantile can be smaller than $m\min\{T, q_*\}/16$. The comparisons $m \ge K/2$ and $\log(m/\delta) \ge \frac{1}{2}\log(K/\delta)$ yield

$$\mathfrak{A}_{T,K} \ge 2^{-24}\min\{KT, K^2\eta^{-2}\log(K/\delta)\}.$$

The lower bound uses a static first-round burst; it does not attribute its statistical logarithm uniquely to dynamic arrivals.

At fixed task parameters, uniformly over schedules satisfying a terminal count limit $K_T \le k(T)$, sublinear expected regret and sublinear area envelopes are simultaneously attainable if and only if $k(T)^2 \log k(T) = o(T)$. Necessity is the last lower bound; sufficiency follows from the displayed upper bounds. This terminal-count-class conclusion allows the worst schedule to depend on T.

A fixed sparse schedule that still defeats a uniform area envelope. Set

$$s_n = 2^{2^n}, \qquad m_n = \left\lfloor \frac{\sqrt{s_n}}{(\log s_n)^{1/4}} \right\rfloor .$$

Release one reference at time one and m_n candidates at each s_n. Earlier batch sizes sum to $o(m_n)$, so $K_{s_n}/\sqrt{s_n} \to 0$; between batches the ratio decreases. Thus $K_t = o(\sqrt{t})$ on this one schedule. For each selected batch n, give the reference mean $1/2 + \varepsilon$, that batch means $1/2 \pm 2\eta$, and other candidates mean $1/2$. Let

$$h_n = \left\lceil \frac{m_n \log(m_n/\delta)}{131072\eta^2} \right\rceil - 1, \qquad T_n = s_n + h_n - 1.$$

For large n, $h_n = o(s_n)$ and the next batch has not arrived. The old history is independent of the new signs. Single-window truncation supplies one fixed configuration for this endpoint with $P(\mathcal{U}_{T_n} \ge m_n h_n/8) \ge 1/7$. But

$$\frac{m_n h_n}{T_n} \asymp_{\eta,\delta} \sqrt{\log s_n} \longrightarrow \infty.$$

No deterministic envelope $a_T = o(T)$ can consequently hold uniformly over reward environments for this schedule. The reward environment may vary with the chosen endpoint; this is exactly the uniform-in-environment statement and is not a claim of simultaneous failure at all endpoints in one fixed reward environment.

S11 A quantile lower bound under the same maintenance protocol

The following result strengthens the expectation lower bound. Its statement uses the same reporting event and the same cumulative unresolved area; it does not introduce a separate deadline requirement. Write $\overline{P}$ for the product prior on all candidate signs in the repeated-update family.

Lemma S11.1 (Completion with a bounded candidate-sampling budget)**.** *Consider one batch of m candidates in its validity window, under the assumptions of the partial-recovery lemma. Let τ be the first round at which at least $m/2$ candidates have determinate reports, or the end of the window if this does not occur. Let S denote that completion occurs, and let $N(\tau)$ count only samples of this batch. Put*

$$L = \frac{m \log(m/\delta)}{1024\kappa}, \qquad q = \lfloor L/2 \rfloor .$$

If $L \geq 4$, then

$$\overline{P}\{S,\ N(\tau) \leq q\} < \frac{1}{2}.$$

Proof. Stop at the earliest of τ, the window endpoint, and the first round with $q+1$ candidate samples. At this bounded stopping time ζ, the number of candidate samples is at most $q+1 < L$. On $\{S, N(\tau) \leq q\}$, the budget stopping condition has not fired before τ, so $\zeta = \tau$ and at least $m/2$ reports are determinate. If the displayed event had probability at least $1/2$, the partial-recovery lemma at ζ would give $\overline{E}N(\zeta) \geq L$, contradicting the deterministic bound $N(\zeta) \leq q+1 < L$.

The extra sample in the stopping rule is needed. Stopping exactly at the qth candidate sample would miss a completion that uses q candidate samples and later observations of the reference arm. □

Theorem S11.2 (Regret quantiles required by rapid maintenance)**.** *In the repeated-update family with J validity windows, M candidates per window, $M \geq \max\{2, J\}$, $Jv \leq \varepsilon/4$, $d \leq v/32$, $\eta = d/2$, and $0 < \delta \leq 1/16$, suppose a policy is δ-reliable on the family and, for every sign configuration σ,*

$$P_\sigma\{E_{\mathrm{lab}},\ \mathcal{U}_T(\eta) \leq JMH/8\} \geq 1 - \delta.$$

Then there is one fixed configuration σ for which

$$P_\sigma\left\{R_T \geq 2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}\right\} \geq \frac{11}{80}.$$

This conclusion permits the policy to know the reference identities, the arrival schedule, and the horizon.

Proof. Let S_e denote completion of at least half of the current candidates before the next reference arrives, and let N_e be that batch's sample count at its first such completion, or at the window endpoint when completion fails. Set

$$L_0 = \frac{M\log(M/\delta)}{32768d^2}, \qquad q_0 = \lfloor L_0/2 \rfloor.$$

The two directional Bernoulli divergences are at most $32d^2$, so the partial-recovery lower bound is at least L_0. The assumptions imply $d \leq 1/1024$ and $M \geq 2$, hence $L_0 > 4$. The preceding budget argument, with $q_0 + 1 < L_0$, gives

$$\overline{P}(F_e) < \frac{1}{2}, \qquad F_e := \{S_e, N_e \leq q_0\}.$$

The argument is valid with all previous-window information included in the history. Those observations and later reference distributions carry no information about this batch's independent signs.

On the area-success event, any window without S_e contributes at least $MH/2$ area. Consequently $\sum_e 1_{S_e} \geq 3J/4$ on an event of prior probability at least $1-\delta$. If $F = \sum_e 1_{F_e}$, then $\overline{E}F \leq J/2$ and

Markov's inequality gives

$$\overline{P}\{F \le 5J/8\} \ge 1/5.$$

No independence across windows is asserted or needed. Intersecting these two events has probability at least $1/5 - \delta \ge 11/80$. On their intersection at least $J/8$ windows complete with $N_e > q_0$, and therefore with $N_e > L_0/2$. Each candidate observation in its birth window has gap at least $\varepsilon - d \ge \varepsilon/2$. These are observations of distinct batches in disjoint windows. Pathwise on that intersection,

$$R_T \ge \frac{\varepsilon}{2}\frac{J}{8}\frac{L_0}{2} = 2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}.$$

A mixture probability at least $11/80$ implies that one fixed configuration has at least that probability. Thus no different worst environment is selected for different windows. □

Corollary S11.3 (Rapid maintenance with a joint regret quantile)**.** *Let $\mathfrak{R}_\Theta^{\mathrm{joint}}(a)$ be the infimum of r over policies that are reliable on the full bounded model and satisfy*

$$\inf_{\theta\in\Theta} P_\theta\{E_{\mathrm{lab}}, \mathcal{U}_T(\eta) \le a, R_T \le r\} \ge 1 - \delta.$$

For the repeated-update horizon $H = 128(M + J + 1)\lceil 32d^{-2}\log\{4K(K+1)/\delta\}\rceil$, where $K = J(M+1)+1$, universal positive constants c, C satisfy

$$c\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta} \le \mathfrak{R}_\Theta^{\mathrm{joint}}(JMH/8) \le C\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}.$$

Proof. For the lower bound, every policy feasible for the joint criterion satisfies the area premise of Theorem S11.2. Since $11/80 > \delta$, a joint regret bound strictly below its threshold is impossible. Taking an infimum gives the claimed lower bound. For the upper bound, the fine-checkpoint policy has the required area bound on its simultaneous interval-coverage event, whose probability is at least $1 - \delta/2$. Independently of coverage, every available arm in the family has gap at most 2ε, so $R_T \le 2\varepsilon T$ on every sample path. The specified horizon and $J \le M$ yield the displayed upper order, with the same universal constant as the rapid-maintenance upper bound. □

S12 A horizon-truncated lower bound across compatible task windows

Proposition S12.1 (Additive truncated area in one compatible experiment)**.** *Consider finitely many batches indexed by e. Batch e contains m_e independent signs and has a deterministic validity window W_e of length H_e that begins at that batch's arrival round. During that window each sign is its arm's relative-goodness label and its boundary distance exceeds η. The signs across all batches are independent, other distributions do not encode them, and no information about batch e is available before its arrival. Samples remain interleaved and adaptive. Suppose every sign configuration has familywise reporting error at most $0 < \delta \le 1/16$. Let $0 < \kappa_e < \infty$ bound the two directional*

divergences in batch e, and define

$$h_e = \min\left\{H_e, \left\lceil \frac{m_e \log(m_e/\delta)}{1024\kappa_e} \right\rceil - 1\right\}, \qquad S = \sum_e m_e h_e.$$

When $S > 0$, one fixed full sign configuration satisfies

$$P\{\mathcal{U}_T(\eta) \geq S/8\} \geq 1/7,$$

provided every selected window lies before T and every candidate–time pair is counted only once. The windows may overlap when they concern distinct arms and their stated qualification structure is jointly realizable.

Proof. At the rth round after batch e arrives, $r \leq h_e$, the entire experiment has supplied at most r samples of that batch. The contrapositive of the partial-recovery lemma shows that at least half the batch is determinate with probability less than $1/2$. Its expected unresolved count is therefore at least $m_e/4$. Let Y sum unresolved candidate–time pairs over the first h_e rounds of all selected windows. All these pairs have boundary distance larger than η, so $Y \leq \mathcal{U}_T(\eta)$, while $0 \leq Y \leq S$ and $\overline{E}Y \geq S/4$. If $p = \overline{P}(Y \geq S/8)$, then $S/4 \leq S/8 + (7S/8)p$, giving $p \geq 1/7$. Select a fixed configuration from the common product prior. Compatibility of the windows is essential: this proof cannot sum bounds from mutually incompatible reference environments. □

S13 A general information-allocation lower relaxation

This result is useful beyond the separated family, but does not assert attainability. Fix a reference environment $\boldsymbol{r}$, endpoint T, and a deterministic query set

$$\mathcal{Q} \subseteq \{(i,t) : i \in \mathcal{A}_t, |\Delta_i(t) - \varepsilon| > \eta\}, \qquad W = |\mathcal{Q}|.$$

For each query choose any allowed alternative $\boldsymbol{r}'$ that reverses arm i's membership at time t. Put $u_{i,t} = P_{\boldsymbol{r}}(i \in \mathcal{V}_t)$ and

$$\Phi_\delta(u) = \begin{cases} \mathrm{kl}(1 - \delta - u, \delta), & 0 \leq u < 1 - 2\delta, \\ 0, & 1 - 2\delta \leq u \leq 1. \end{cases}$$

This function is convex and nonincreasing: the positive branch is an affine composition of binary KL, and its value and derivative vanish at the join.

Proposition S13.1 (One protocol for area and information)**.** *For any reliable policy,*

$$\sum_j E_{\boldsymbol{r}} N_j(t)\, \mathrm{kl}(r_j, r'_j) \geq \Phi_\delta(u_{i,t}).$$

If its joint area budget at $\boldsymbol{r}$ is a, then $\sum_{(i,t)\in\mathcal{Q}} u_{i,t} \leq a + \delta W$. Consequently $E_{\boldsymbol{r}} R_T$ is at least the

value of

$$\inf_{x,u} \sum_{s\le T} \sum_{j\in\mathcal{A}_s} \Delta_j(s) x_{s,j},$$

subject to $x_{s,j} \ge 0$, $\sum_{j\in\mathcal{A}_s} x_{s,j} = 1$, *no prearrival allocations, the preceding unresolved-probability constraint, and for every query and chosen alternative,*

$$\sum_{s\le t} \sum_{j} x_{s,j} \,\mathrm{kl}(r_j, r'_j) \ge \Phi_\delta(u_{i,t}).$$

For non-Bernoulli laws replace binary arm divergences by their distributional KL.

Proof. Let F be the event of issuing, at (i,t), the determinate label correct in the reference environment. Then $P_{\boldsymbol{r}}(F) \ge 1 - u_{i,t} - \delta$, whereas $P_{\boldsymbol{r}'}(F) \le \delta$. The history-KL identity and data processing give the claimed constraint whenever $1 - u_{i,t} - \delta > \delta$; otherwise nonnegativity suffices. This is the precise point where the truncated KL is needed. On area success the queried unresolved sum is at most a; outside success it is at most W. Taking expectations gives $a + \delta W$. The actual occupation probabilities $x_{s,j} = P_{\boldsymbol{r}}(I_s = j)$ satisfy every constraint, with objective exactly $E_{\boldsymbol{r}} R_T$. □

One observation can enter many information constraints but is charged once in the objective. Neither convex feasibility nor an oracle allocation proves that a policy with unknown means and future arrivals attains this relaxation. Dropping δW would also change the stated risk protocol.

S14 Intermediate configurations and known-reference diagnostics

In the homogeneous repeated-update family, define

$$B_{J,M} = 2(M+1)\{J(2M+1) + J(J-1)/2\} + 2(M+J+1).$$

If a predetermined set of s of the J candidate batches uses precision $\rho_f = d/4$ and the rest uses $\rho_c = v/8$, the shared-evidence bound gives

$$\mathcal{U}_T(\eta) \le (J-s)MH + B_{J,M} q_f$$

on checkpoint coverage. The unconditional reward bound gives

$$ER_T \le 2\varepsilon M\{sq_f + (J-s)q_c\} + 64\left(\frac{JM}{\varepsilon} + \frac{J^2}{v}\right)\log(2T) + 1.$$

Every adjacent precision pair satisfies $d + \rho_e + \rho_{e+1} < v$, and every window completes its checkpoints, so the proof follows by applying the same batchwise accounting. The choice of which batches receive fine service must be made without their reward data. These are attainable points, not a proof of optimality at every area budget.

An additional diagnostic gives each batch's reference identity to the policy and discloses that its role is best within its own window. Its numerical mean remains unknown. Sample the reference

Table 1: Allocation diagnostics. Each cell gives mean forced service pulls and mean final η-unresolved arms. UCB and MOSS have zero forced service.

Method	Stable	Improvements	Burst	Stop
CP-FIFO ($\rho = .04$)	10000, 8.00	10000, 6.00	10000, 5.00	10000, 5.00
CP-FIFO ($\rho = .12$)	7340, 0.16	5786, 0.00	5991, 0.00	5737, 0.00
CP-confidence	7144, 0.16	4653, 0.00	6092, 0.00	5322, 0.00
CP-period8	2500, 6.74	2500, 0.20	2225, 3.00	2500, 2.00
UCB-checkpoint	0, 8.00	0, 3.00	0, 4.00	0, 4.00
UCB-prefix	0, 6.23	0, 0.11	0, 1.69	0, 1.71
Growing-MOSS	0, 7.00	0, 1.00	0, 4.00	0, 4.00
ALL-ε adapt.	3696, 0.00	10000, 0.00	7794, 0.00	10000, 0.00
Batch100-UCB	0, 6.40	0, 0.53	0, 4.00	0, 2.89

to its checkpoint, then its candidates to theirs, and otherwise sample the latest reference. Keep the same reporting comparisons. For windows long enough to complete this work, the regret in the specified family is pathwise

$$R_T = \sum_{e,k} (\varepsilon - \sigma_{e,k} d) q_{e,k}.$$

Reference sampling and idle exploitation have zero regret, whereas each candidate checkpoint occurs in its birth window and pays its displayed gap. The coarse version has $R_T \leq 2\varepsilon J M q_c$ with no extra $\log T$ reward term. Comparing with the rapid lower bound isolates a squared v/d cost ratio. This additional role information makes the allocation problem easier; this policy is not the main unknown-reference algorithm. Its interval-based label reliability still holds on the full bounded model, while its reward statement only holds on the disclosed-role performance family.

S15 Additional computational details

The main reporting-comparison table reports six methods; its cost-area figure includes all nine. These reporting experiments concern CP-UCB-AA and its comparators, not the repaired two-stage UCB-AA procedure. The allocation table below records every service variant. The full configuration lists each arm's arrival time and mean, the shared potential reward construction, fixed policy parameters, and seeds. The actual final arm count and endpoint are used by the evaluator, not supplied to a policy. All methods receive the same observed-history information. The natural ALL-ε comparator is fully specified by the released code and is not claimed to reproduce FAREAST or $(\mathrm{ST})^2$.

The compiled CP reference was compared to its original public class for 20,000 successive steps, checking choices, service flags, counts and reports. The experiment records include per-run coverage and label errors, checkpoint completion, service regret, and the three-part regret identity. Their largest decomposition discrepancy was below 3.1×10^{-12}. These checks validate the numerical implementation and accounting; they are not formal probability certificates or proofs of the analytical lower bounds.

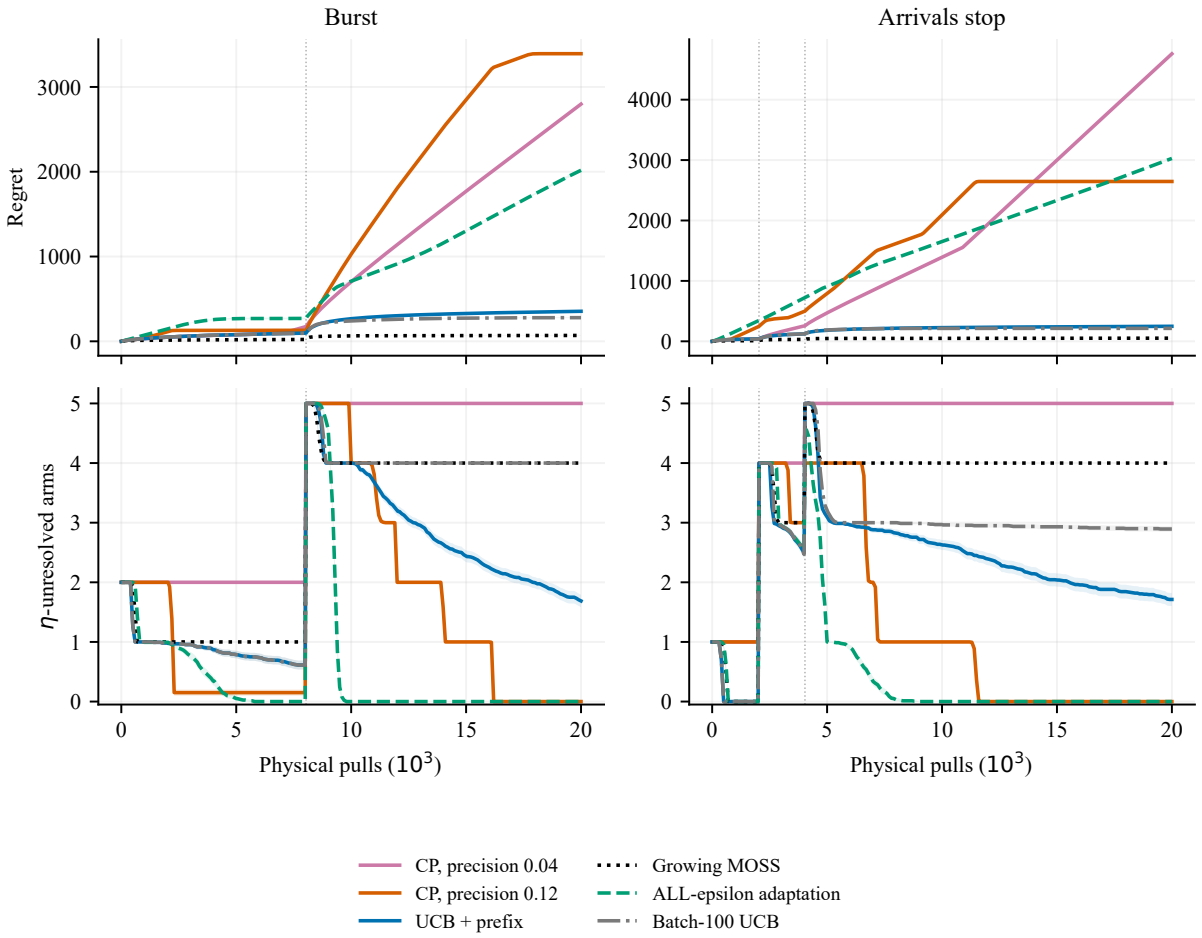


Figure 1: Additional paths for the burst and stopped-arrival environments. The displayed policies, common potential reward streams, 160 replications, and 1.96 Monte Carlo standard error bands are the same as in the main article. Vertical lines mark actual arrivals. The full run-level data retain all nine policies.

S16 Preserved figures from the original simulation study

The original progressive step-up and near-tied top-set experiments are preserved as an archived implementation record. Each scenario has 100 replications at $T = 60{,}000$. The archived UCB-AA uses $M = 5$ in the first scenario and $M = 4$ in the second, then commits to a fixed survivor pool. The source snapshots and endpoint data show that later buffered arms are not necessarily admitted. Its all-history empirical means and incumbent reference comparisons also differ from the repaired prefix procedure. The figures below therefore retain the original observations without claiming validation of the revised fixed-budget algorithm.

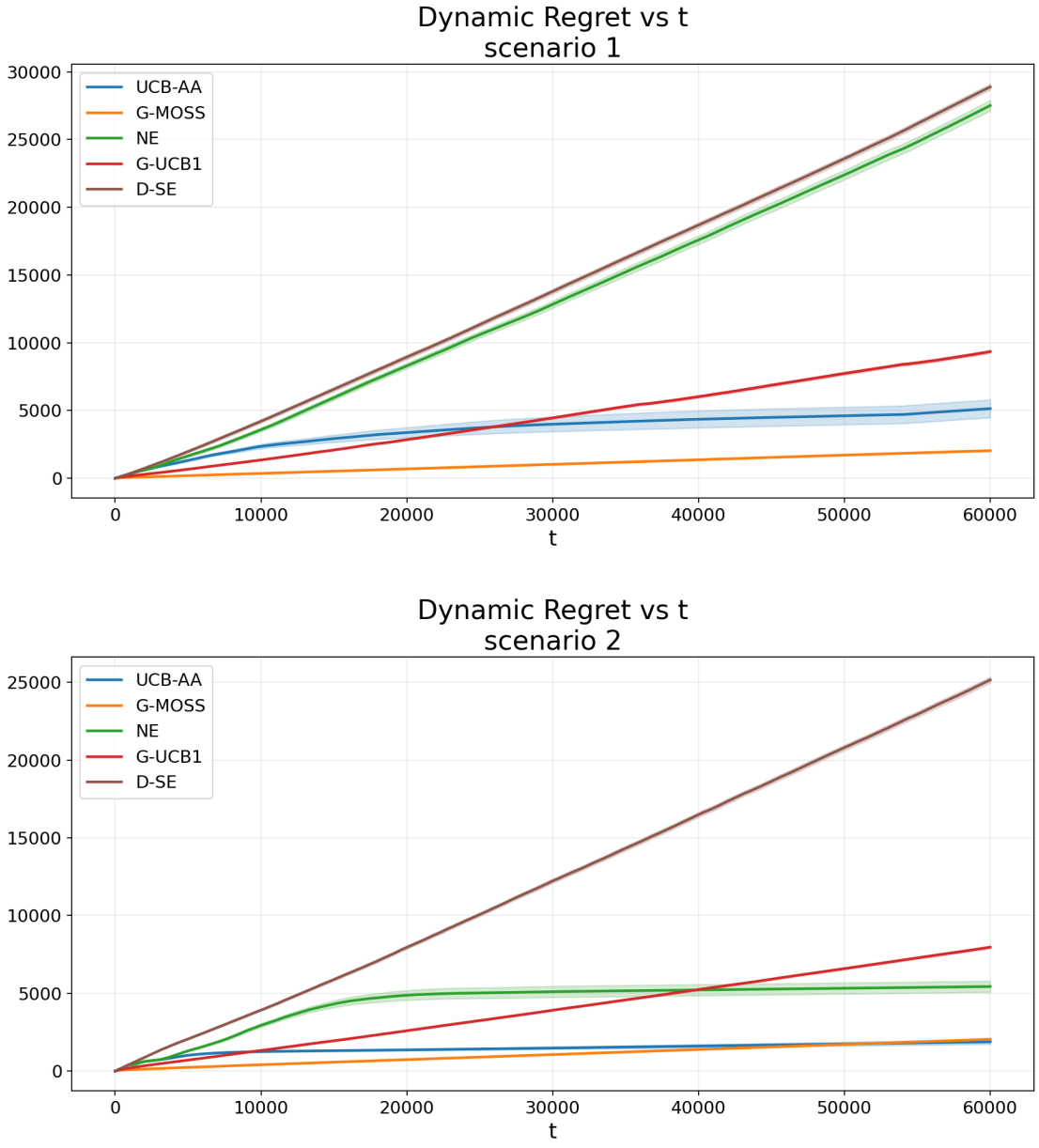


Figure 2: Preserved cumulative dynamic regret curves from the original archived implementation. The step-up scenario is above and the near-tied scenario below. The original uncertainty bands and 100 replications are retained. The archived current-time MOSS index and finite-round UCB-AA behavior are specified in the released source snapshots.

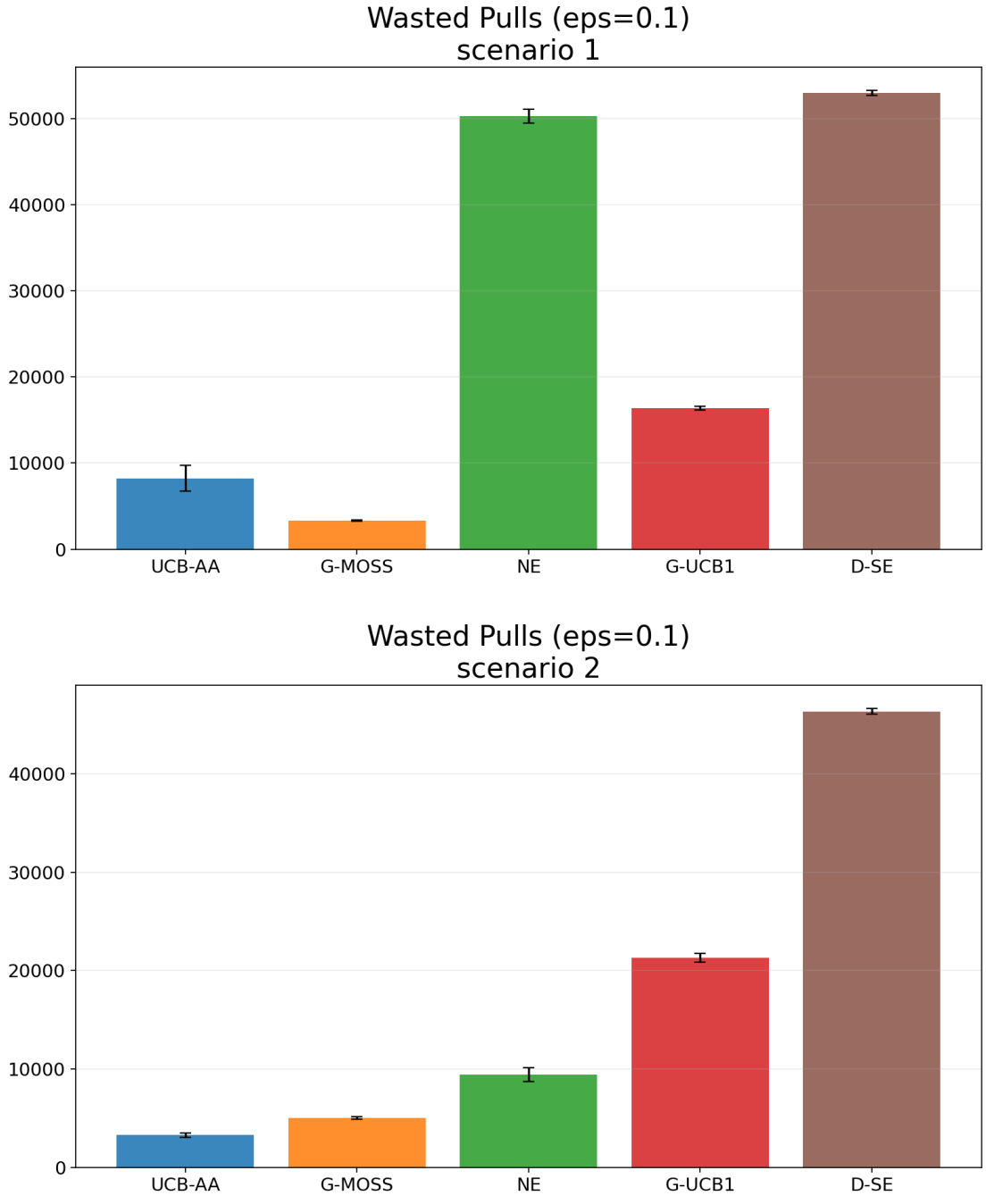


Figure 3: Preserved wasted-pull comparisons from the original step-up and near-tied scenarios, with threshold 0.1. These are archived implementation results, not new simulations of the repaired algorithm.

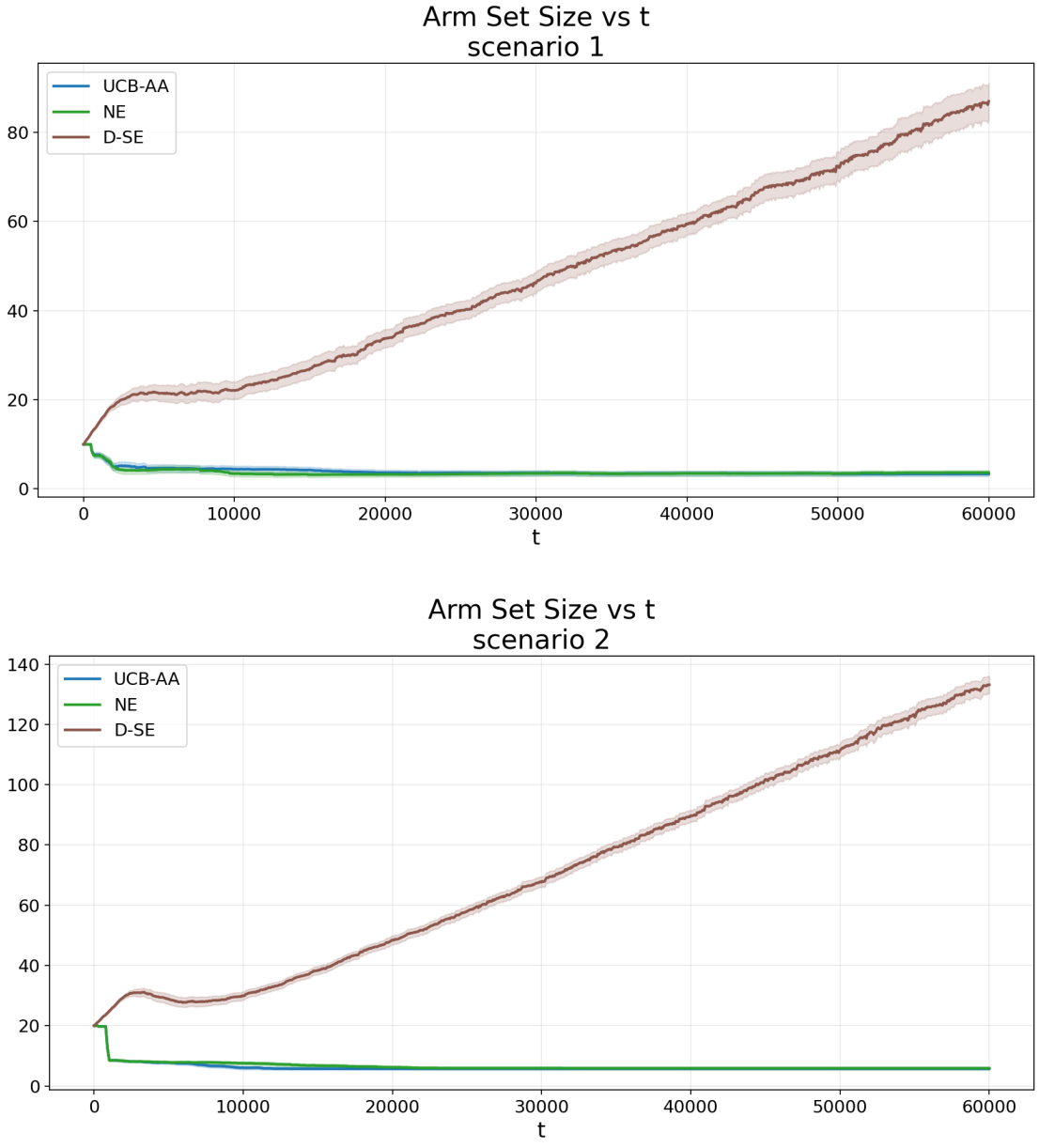


Figure 4: Preserved original active-set curves. A smaller active set does not imply that every later arrival was statistically excluded: the archived UCB-AA implementation stops admission after its prescribed rounds.

References

Kevin Jamieson and Lalit Jain. A bandit approach to sequential experimental design with false discovery control. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Emilie Kaufmann, Olivier Cappé, and Aurélien Garivier. On the complexity of best-arm identification in multi-armed bandit models. *Journal of Machine Learning Research*, 17(1):1–42, 2016.

Matthew L. Malloy and Robert D. Nowak. Sequential testing for sparse recovery. *IEEE Transactions on Information Theory*, 60(12):7862–7873, 2014.

Blake Mason, Lalit Jain, Ardhendu Tripathy, and Robert Nowak. Finding all ε-good arms in stochastic bandits. In *Advances in Neural Information Processing Systems*, volume 33, 2020.